

7. Write the python program to implement BFS.

Aim

To develop a Python program to implement the Breadth-First Search (BFS) algorithm for traversing a graph starting from a user-specified source node.

Algorithm

1. Start the program.
 2. Read the number of vertices and edges from the user.
 3. Create an adjacency list to represent the graph.
 4. Read all the edges and store them in the adjacency list.
 5. Read the starting vertex.
 6. Initialize an empty queue and a visited set.
 7. Insert the starting vertex into the queue and mark it as visited.
 8. Repeat until the queue becomes empty:
 - Remove the front vertex from the queue.
 - Display the current vertex.
 - Visit all adjacent unvisited vertices.
 - Mark them as visited and insert them into the queue.
 9. Stop.
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Program

```
from collections import deque

# Read number of vertices and edges
n = int(input("Enter the number of vertices: "))
e = int(input("Enter the number of edges: "))

# Create adjacency list
graph = {}

for i in range(n):
    vertex = input(f"Enter vertex {i+1}: ")
```

```

graph[vertex] = []

# Read edges
print("\nEnter the edges:")
for i in range(e):
    u, v = input().split()
    graph[u].append(v)
    graph[v].append(u)          # Remove this line for directed graph

# Read starting vertex
start = input("\nEnter the starting vertex: ")

# BFS Traversal
visited = set()
queue = deque()

visited.add(start)
queue.append(start)

print("\nBFS Traversal:")

while queue:
    vertex = queue.popleft()
    print(vertex, end=" ")

    for neighbor in graph[vertex]:
        if neighbor not in visited:
            visited.add(neighbor)
            queue.append(neighbor)

```

Input

Enter the number of vertices: 5

Enter the number of edges: 5

Enter vertex 1: A
Enter vertex 2: B
Enter vertex 3: C
Enter vertex 4: D
Enter vertex 5: E

Enter the edges:

A B

A C

B D

C E

D E

Enter the starting vertex: A

Output

```
BFS Traversal:  
>>> C D E
```

Result

The Python program was successfully implemented to perform Breadth-First Search (BFS) on a graph. The program accepts the graph and starting vertex from the user, traverses the graph level by level using a queue, and displays the BFS traversal order.